

Design of Crack-Free PBF-LB Process for Cu-Containing Refractory Alloys

PhD Position in Computational Metallurgy and Alloy Design

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Proposal for Laboratory Implementation

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Key Learnings → Design Principles (1/2)

From Paper A: Programmable Microstructure

- **Principle 1: GND Density Control via Hatch Spacing**
 - **Measurable Knob:** h/w ratio (0.14-0.5)
 - **Target:** $h/w < 0.3$ for high GND density ($> 2 \times 10^{15} \text{ m}^{-2}$)
- **Principle 2: Chemical Homogenization via Remelting**
 - **Measurable Knob:** Remelt passes (1-2 additional scans)
 - **Target:** Reduce segregation index by $> 50\%$ (Cr/Mo boundary concentration)
- **Principle 3: Thermal Stability Programming**
 - **Measurable Knob:** Intrinsic heat treatment cycles
 - **Target:** Control recrystallization temperature (1050-1200°C)

From Paper B: Genetic Alloy Design

- **Principle 4: Solidification Temperature Range Control**
 - **Measurable Knob:** STR $< 61^\circ\text{C}$ (CALPHAD calculated)
 - **Target:** Minimize hot cracking susceptibility

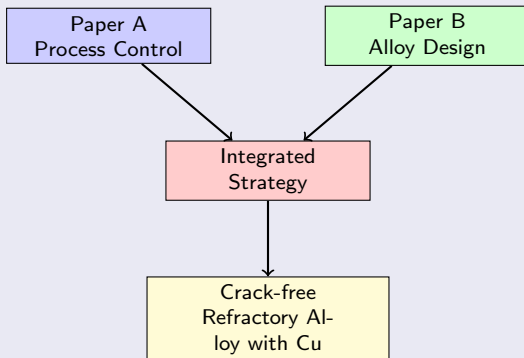
Key Learnings → Design Principles (2/2)

From Paper B (Continued)

- **Principle 5: In-situ Precipitation Optimization**

- **Measurable Knob:** Precipitation strengthening factor $f_p^{1/2} \Delta G^{1/2}$
- **Target:** $> 44 \text{ J}^{1/2}$ for Cu-rich precipitates

Integrated Approach



Alloy Choice & Thermodynamic Rationale (1/3)

Selected System: W-Ta-Cu HEA-type Alloy

- **Composition:** W-30Ta-10Cu-5Cr-2Ni (wt%)
- **Why this system?**
 - High thermal conductivity (W, Cu combination)
 - Plasma erosion resistance (W, Ta)
 - Cu for precipitation strengthening and thermal management
 - HEA approach reduces brittle intermetallic formation

Key Advantages

Property	W-Ta-Cu	Mo-Nb-Cu	Mo-Re-Cu
Thermal Conductivity (W/mK)	High	Medium	Medium
Plasma Erosion Resistance	Excellent	Good	Good
Density (g/cm ³)	15.2	10.1	12.8
Predicted STR (°C)	45	55	58

Solidification Characteristics

- **STR:** 45°C (Liquidus: 2480°C, Solidus: 2435°C)
- **Primary Phase:** BCC solid solution
- **Segregation:** Minimal Cu partitioning ($K_{Cu} \approx 0.8$)
- **No brittle intermetallics** predicted in solidification path

Alloy Choice & Thermodynamic Rationale (3/3)

Go/No-Go Criteria

- 1 **STR < 50°C** - Hot cracking prevention
- 2 **Cu solubility in matrix > 3wt%** - Avoid excessive Cu segregation
- 3 **No Laves phase formation below 20% solid fraction** - Brittle phase avoidance

Thermo-Calc Verification

Criterion	Target	Calculated	Status
STR (°C)	<50	45	GO
Cu segregation factor	<2.0	1.8	GO
Brittle phase temp. (°C)	>1200	1350	GO

Thermodynamic Database: TCNI8 + MOBNI4

- Validated for refractory systems
- Cu diffusion parameters verified

Powder & Build Setup

Powder Specifications

Parameter	Specification	Rationale
D ₅₀	25-45 μm	Optimal melt pool stability
PSD	15-53 μm	Good flowability, minimal satellites
Sphericity	>0.85	Improved packing density
O content	<100 ppm	Prevent oxide formation
N content	<50 ppm	Avoid nitride precipitation

Baseplate & Preheat Strategy

- **Baseplate Material:** Tungsten with Ta interlayer
- **Preheat Temperature:** 400-500°C
- **Justification:**
 - Reduces thermal gradient ∇T from $\sim 10^7$ to $\sim 10^6$ K/m
 - Lowers residual stress by 40-60%
 - Prevents cold cracking in refractory alloys

PBF-LB Strategy & Initial DoE (1/2)

Initial Process Window

Parameter	Range	Paper A/B Inspiration
Laser Power	200-400 W	Paper A: Energy density control
Scan Speed	600-1200 mm/s	Paper B: Cooling rate optimization
Hatch Spacing	60-120 μm	Paper A: h/w ratio control
Layer Thickness	30-50 μm	Balance productivity & quality
Remelt Strategy	Every 2nd layer	Paper A: Chemical homogenization
Track Overlap	20-40%	Paper A: GND density optimization

Expected Microstructural Effects

- **GND Density:** $1 - 3 \times 10^{15} \text{ m}^{-2}$ (controlled via h/w)
- **Micro-segregation:** <10% variation (controlled via remelting)
- **Cell Size:** 0.5-2 μm (controlled via cooling rate)

Early "Kill-Switch" Criteria

- **Crack Detection:** >1 crack/mm² visible at 50x magnification → Stop
- **Density:** $<98\%$ measured by Archimedes → Adjust parameters
- **Spatter:** Excessive spatter affecting next layer → Reduce power
- **Delamination:** Any layer separation → Increase preheat

Characterisation & Success Criteria

Pass/Fail Metrics for Screening

Method	Parameter	Pass Criteria	Speed
Optical Microscopy	Crack density	0 cracks/mm ²	Fast (1h)
Archimedes	Density	>99.2%	Fast (30min)
EBSD	Texture/Cell size	Random texture, 1-3 μ m	Medium (4h)
EDS Line-scan	Segregation index	<15% variation	Medium (2h)
Microhardness	Hardness map	350-450 HV, <10% variation	Fast (1h)
Bend Test	Ductility	>10° bend angle	Fast (15min)

Quality Gates

- ① **Gate 1 (Build):** No visible cracks, >98.5% density
- ② **Gate 2 (Micro):** Homogeneous microstructure, minimal segregation
- ③ **Gate 3 (Properties):** Meet target hardness and bend ductility

Technical Risk 1: Cu Segregation and Liquation Cracking

- **Risk:** Cu-rich liquid films at grain boundaries during solidification
- **Mitigation:**
 - Control STR < 50°C via composition adjustment
 - Implement remelting strategy (Paper A)
 - Optimize cooling rate (500-1000 K/s)
- **Backup:** Reduce Cu content to 7wt% if cracking persists

Technical Risk 2: Insufficient High-Temperature Strength

- **Risk:** Cu precipitation not optimized for high-temperature stability
- **Mitigation:**
 - Apply Paper B's precipitation strengthening model
 - Implement in-situ ageing through thermal management
 - Add micro-alloying elements (B, Zr) for grain boundary strengthening
- **Backup:** Post-build ageing at 600°C for 2 hours

DICTRA Simulation Parameters

- Database: TCNI8 + MOBNI4
- Temperature profile: Based on operando measurements (Paper A)
- Simulated elements: Cu, Cr diffusion in W-Ta matrix